Andhra University College of Engineering

Information Technology | Course Code: IT301

Web Technologies & Architecture - Laboratory Manual & Practical Guide

Syllabus Focus: IT301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Semantic Web & Modern Responsive CSS

" Syllabus Scope: HTML5 semantic standards, CSS Flexbox and Grid layouts, Media queries, Web accessibilit

Unit 2: JavaScript Engine & Asynchronous Architecture

" Syllabus Scope: ECMAScript 6+ specifications, Closures, Prototypal inheritance, Promises, Async/Await,

IT301: Web Technologies & Architecture - Advanced Topics & Assessment

Academic Track: Information Technology - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Server Middleware & RESTful API Design

" HTTP/1.1 vs HTTP/2 protocol, Node.js runtime, Express/Next.js middleware, REST architecture, JSON serializat

Unit 4: State Management & Client-Server Auth

" Cookies, LocalStorage, Session tokens, RFC 7519 JSON Web Tokens JWT, Password hashing, Role-based access con

Unit 5: Web Security & Contemporary Architectures

" XSS, CSRF, SQL Injection, Content Security Policy CSP, CORS, Service Workers, Progressive Web Apps PWA

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Building a Responsive Multi-Column Layout using CSS Flexbox and CSS Grid without Frameworks
Lab 2: Dynamic DOM Manipulation and Asynchronous Fetch API Requests with Error Handling
Lab 3: Implementation of a RESTful Backend API with JWT Authentication and Scrypt Password Hashing
Lab 4: Automated Testing of REST API Endpoints using Integration Test Suites

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* David Flanagan - JavaScript: The Definitive Guide O Reilly 7th Ed
* Jon Duckett - HTML and CSS: Design and Build Websites Wiley

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